PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

[1] Only one or two problems on each assignment are marked for submission and grading.

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4.4.5 Heat Capacities from Equipartition

6 Entropy

As previously mentioned, Sadi Carnot was the first to recognize that heat flowing from hot to cold can be used to produce work, but never all of it. The explanation behind this limit is entropy. The second law of thermodynamics can be restated in terms of entropy:

Law 1 (The Second Law of Thermodynamics (Entropy Formulation))**.** *the entropy of an isolated system never decreases with time. As a result, isolated systems evolve toward a thermodynamic equilibrium state where the entropy is maximized.*

6.1 The Clausius Inequality

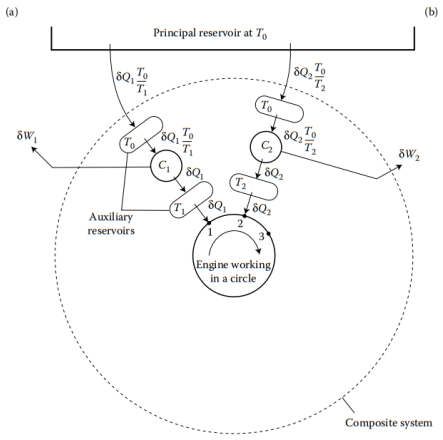


Figure 1: Clausius' thought experiment: Carnot engines route all heat exchanged by the gas through a single reservoir at T_0.

Could an engine run on heat drawn from the surrounding air alone, without burning fuel?

Remark 1 (Clausius' Thought Experiment)**.** *Consider Figure 1. Heat* $\delta Q_1 \frac{T_0}{T_1}$ *flows from the principal reservoir (PR) at* T_0 *(e.g. the surrounding air) to an auxiliary reservoir at* T_0*. A Carnot engine takes this heat, transforms part of it into work, and transfers the rest,* δQ_1*, to an auxiliary reservoir at the gas's temperature* T_1*, which passes it to the gas, changing its state from* (P_1, V_1, T_1) *to* (P_2, V_2, T_2)*. Each auxiliary reservoir gives exactly what it receives, so it ends unchanged. This continues until the gas is back at its original state. The question is whether heat from a single reservoir can thus be fully converted into work, with no other effect on the surroundings.*

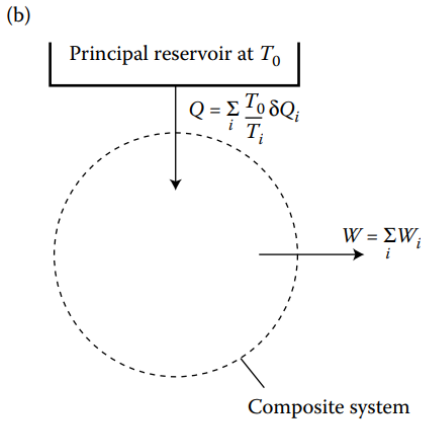


Figure 2: The composite system exchanges heat only with the reservoir at T_0.

After a full cycle, everything in the composite system (Figure 2) is back to its original state, so $\Delta U = 0$ and the first law gives

$$W = Q = T_0 \sum_i \frac{\delta Q_i}{T_i}, \tag{1}$$

where $W = \sum_i \delta W_i$ is the total work done and Q the heat drawn from T_0. A positive $W = Q$ is forbidden by:

Law 2 (Kelvin Statement of the Second Law of Thermodynamics)**.** *No process is possible whose sole result is the absorption of heat from a reservoir and the conversion of this heat into work.*

Hence $W = Q \leq 0$, and since $T_0 > 0$, taking the steps infinitesimal gives:

Law 3 (Clausius Inequality)**.** *For any cyclic process,*

$$\oint \frac{\delta Q}{T_0} \leq 0, \tag{2}$$

where T_0[2] *is the temperature, of the auxiliary reservoirs supplying heat to the working substance, at which* δQ *is supplied*[3]*, with equality if and only if the cycle is reversible. If the you run the cycle in reverse, then*

$$\oint \frac{\delta Q}{T_0} \geq 0. \tag{3}$$

[2]The integral does not depend on T_i.

[3]The 'surr' can be dropped because for a fully reversible process $T_i = T_{sys} = T_{surr} = T_0$ (no finite difference in T at any point over the cycle).

Remark 2. *Physically, an engine needs a hot source and a colder sink: a car takes heat from burning fuel, turns part of it into work, and dumps the rest into the colder air. With only one reservoir, there is nowhere to dump heat, so no net work is possible.*

6.2 Entropy

Suppose a system is taken along a reversible path Route_1 from an initial state i to a final state f and then back again to the initial state along another reversible path Route_2, completing a reversible cycle. Since the cycle is reversible, the Clausius inequality holds with equality:

$$\oint \frac{\delta Q_{\text{Route}}}{T} = \int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} + \int_{f,\ \text{Route}_2}^{i} \frac{\delta Q_{\text{Route}}}{T} = 0,$$

therefore

$$\int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} = -\int_{f,\ \text{Route}_2}^{i} \frac{\delta Q_{\text{Route}}}{T}.$$

Since Route_2 is reversible, traversing it from i to f instead flips the sign of every δQ, so $\int_{f,\ \text{Route}_2}^{i} = -\int_{i,\ \text{Route}_2}^{f}$ and

$$\int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} = \int_{i,\ \text{Route}_2}^{f} \frac{\delta Q_{\text{Route}}}{T}.$$

The integral is the same along any reversible path between i and f, so it is path independent (depends only on the end states) and defines a state function.

Definition 1 (Entropy)**.** *The entropy S is the state function defined by*

$$\Delta S = S_f - S_i = \int_i^f \frac{\delta Q_{rev}}{T}, \tag{4}$$

where the integral is taken along any reversible path, for which $T_{sys} = T_{surr}$. Only entropy differences (ΔS) are defined, in units of J/K. For an infinitesimal reversible process

$$dS = \frac{\delta Q_{rev}}{T}. \tag{5}$$

Remark 3. *Since S is a state function, ΔS for an irreversible process is computed along any reversible path between the same end states.*

Remark 4. *Water heated from $T_i = 20^\circ C$ to $T_f = 100^\circ C$ on a stove at $100^\circ C$ is irreversible (finite temperature difference). Along a reversible path, $\delta Q_{rev} = C_P\, dT$, so*

$$\Delta S = \int_{T_i}^{T_f} \frac{C_P\, dT}{T} = C_P \ln \frac{T_f}{T_i}, \tag{6}$$

with temperatures in kelvin.

6.3 The Principle of Increasing Entropy

6.3.1 Isolated Systems

6.3.2 Entropy of the Universe

6.3.3 Entropy Maximization and the Arrow of Time

6.4 Entropy–Temperature Diagrams

6.5 The Thermodynamic Identity

6.6 Other Examples of Entropy Calculations

6.6.1 Free Expansion

6.6.2 Measuring Entropies

7 Statistical Mechanics

7.1 Introduction to Probability and Statistics

Definition 2 (Microstate)**.** *A microstate is a specific configuration of the particles in a system.*

Definition 3 (Macrostate)**.** *A macrostate is a description of the system in terms of macroscopic properties, such as temperature, pressure, and volume. Different macrostates can be distinguished by a measurement, while different microstates of the same macrostate cannot.*

Assuming all microstates are equally probable, the probability of any macrostate is given by the multiplicity of that macrostate divided by the sum of all the multiplicities:

$$P(n) = \frac{\Omega(n)}{\sum_i \Omega(i)}. \tag{7}$$

For N two-state particles, the multiplicity $\Omega(N, n)$ of the macrostate with n out of the N particles in one of the two states can be calculated using the binomial coefficient:

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{n!(N-n)!}. \tag{8}$$

There are $N + 1$ macrostates and 2^N microstates in total:

$$\sum_{n=0}^{N} \binom{N}{n} = 2^N. \tag{9}$$

The distribution has mean $N/2$ and standard deviation $\sigma = \sqrt{N}/2$, so its relative width is

$$\frac{\sigma}{N} = \frac{1}{2\sqrt{N}}, \tag{10}$$

meaning the multiplicity becomes sharply peaked at $n = N/2$ as N grows.

For $N \gg 1$, factorials are evaluated with Stirling's approximation:

$$N! \approx N^N e^{-N} \sqrt{2\pi N}, \qquad \ln N! \approx N \ln N - N. \tag{11}$$

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